THE UNIVERSITY OF AZAD JAMMU AND KASHMIR, MUZAFFARABAD

Department of Software Engineering

Lab 01: Use Case Modeling & Requirement Engineering

Course: Software Construction & Development (Lab)
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Question 1: Online Ticket Reservation System

1.1 System Domain & Functional Overview

An **Online Ticket Reservation System** is a computerized platform designed to facilitate ticket inquiries, seat selection, financial clearance, and ticket delivery for travel or entertainment services (bus, train, airline, cinema). The system automates interactions between passengers, administrative authorities, and third-party financial institutions.

1.2 Identified Actors

Actor Name	Actor Type	Role & System Responsibility
Customer / Passenger	Primary Human Actor	Browses trips, inspects seating diagrams, selects seats, reserves tickets, makes online payments, enters promo vouchers, and cancels bookings.
System Administrator	Secondary / Admin Actor	Maintains routes, timings, fares, fleet/seat configurations, oversees user profiles, and audits analytical reports.
Payment Gateway	External System Actor	Validates credit/debit card credentials, checks financial balance, authorizes deductions, and

Actor Name	Actor Type	Role & System Responsibility
		disburses refund transactions.
Notification Service	External System Actor	Automated SMS and email service that generates and delivers real-time booking confirmations and e-tickets.

1.3 System Use Cases & Relationship Analysis

- **Register / Login Account:** Allows user authentication and profile management.
- **Search Trips / View Schedules:** Filters travel routes by date, location, and vehicle category.
- **Select Seat(s):** Interactive visual seat selection. Included in *Book / Reserve Ticket* <<include>> .
- **Book / Reserve Ticket:** Core business use case. Includes *Select Seat(s)* and *Make Online Payment* <<include>> .
- **Make Online Payment:** Collects payment via card or web banking. Includes *Send Booking Confirmation* <<include>> .
- **Apply Promo Code / Discount:** Optional financial discount voucher. Extends *Make Online Payment* <<extend>> .
- **View & Download E-Ticket:** Retrieves digital ticket with boarding details and QR code.
- **Cancel Reservation:** Reverts booking status; automatically includes *Process Refund Transaction* <<include>> .
- **Process Refund Transaction:** Interfaces with Payment Gateway to return eligible fare to customer.
- **Manage Schedules, Reports & Users:** Handled by the System Administrator.

1.4 UML Use Case Diagram

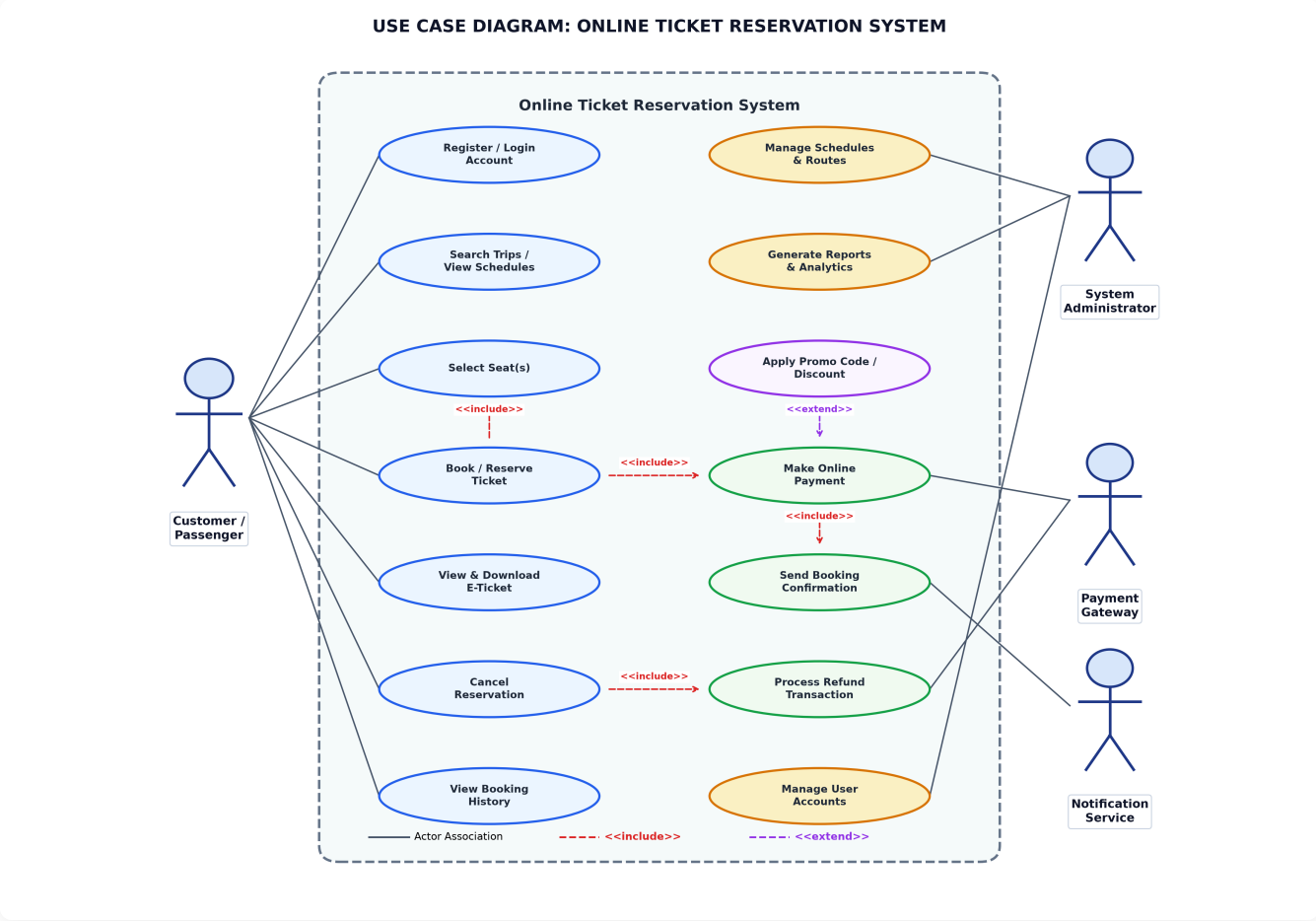


Figure 1.1: Standard UML Use Case Diagram for Online Ticket Reservation System

1.5 Fully Dressed Use Case Specification: Book / Reserve Ticket

Use Case ID & Name	**UC-04: Book / Reserve Ticket**
Primary Actor	Customer / Passenger
Secondary Actors	Payment Gateway, Notification Service
Pre-conditions	Customer is on the trip schedule page with available seats.
Post-conditions	Seats marked as 'Booked', PNR generated, fare debited, e-ticket issued via SMS/email.
Main Success Scenario	1. Customer picks desired trip and initiates booking. 2. System triggers *Select Seat(s)* (<<include>>). 3. Customer selects seats and inputs passenger particulars. 4. System calculates total fare and requests payment.

	5. System invokes *Make Online Payment* (<<include>>). 6. Payment Gateway validates details and authorizes fund deduction. 7. System confirms booking and issues PNR. 8. System triggers *Send Booking Confirmation* (<<include>>). 9. Digital e-ticket is displayed for download.
Alternative Flows	**4a. Promo Code:** Customer enters valid code; system reduces total before payment (<<extend>>). **6a. Payment Failure:** Gateway declines transaction; seats kept locked for 5 minutes for retry.

Question 2: Credit Card Processing System (Case Study)

> ***Problem Statement Case Study:***
> *"The Credit Card Processing System is used to purchase an item from any shop mall and it is used to maintain the limitation of credit card balance and the current transaction process could be updated via a credit card machine. The customer should select the item to be purchased from the shop by using a credit card payment then the vendor should give a bill for the selected item. The customer should give his card to swap and request for the kind of amount transaction. After processing the transaction, the CREDIT CARD MACHINE should give the balance print statement or receipt.*
> *• Customer should select the item from the shop.*
> *• Vendor makes the bill for the selected item.*
> *• Customer gives the credit card to the vendor to swap the card.*
> *• The required amount transaction is done by the card reader.*
> *• Vendor will issue the balance statement to the customer.*
> *• Customers put the signature in the receipt and return to the vendor"*

2.1 Extraction & Identification of Actors

Actor Name	Type	Role & Description in Case Study
Customer	Primary Human	Selects items in the mall, provides credit card for payment, verifies the printed statement, signs the merchant receipt, and returns it.
Vendor / Cashier	Primary Human	Generates the item bill, receives credit card, swipes it in the card reader, initiates amount request, hands receipt to customer, and files the signed receipt copy.
Credit Card Machine / Reader	External Device / Hardware	The physical POS terminal that reads magnetic stripe/chip data, communicates transaction packets, and prints balance statement receipts.
Banking Network / Issuer Bank	Secondary Backend System	Maintains cardholder credit balance limit, validates credit limits against the bill, authorizes or declines transactions, and updates ledger balances.

2.2 Identification of Use Cases

Use Case ID	Use Case Name	Detailed Description
CC-UC-01	Select Item from Shop	Customer chooses mall goods/merchandise to purchase.
CC-UC-02	Make Bill for Selected Item	Vendor calculates total item cost and generates the purchase invoice.
CC-UC-03	Swipe / Insert Credit Card	Customer hands card to vendor; vendor swipes card in reader to capture credentials.
CC-UC-04	Request Amount Transaction	Vendor specifies the billing amount on terminal and submits the transaction.
CC-UC-05	Process Card Transaction	Core transaction processing uniting machine validation and bank authorization.
CC-UC-06	Check & Verify Credit Balance Limit	Bank verifies that the requested purchase amount does not exceed customer's credit limit.
CC-UC-07	Update Account Balance & Record Transaction	Deducts the authorized charge from remaining credit balance and creates transaction log.
CC-UC-08	Handle Declined Transaction	Exception handling executed if credit limit is exceeded or card is invalid (<<extend>>).
CC-UC-09	Print Balance Statement & Receipt	Card machine prints customer and merchant slips showing transaction summary and remaining limit.
CC-UC-10	Sign & Return Receipt to Vendor	Customer endorses merchant copy with signature and returns it to cashier for record keeping.

2.3 Use Case Diagram (Credit Card Processing System)

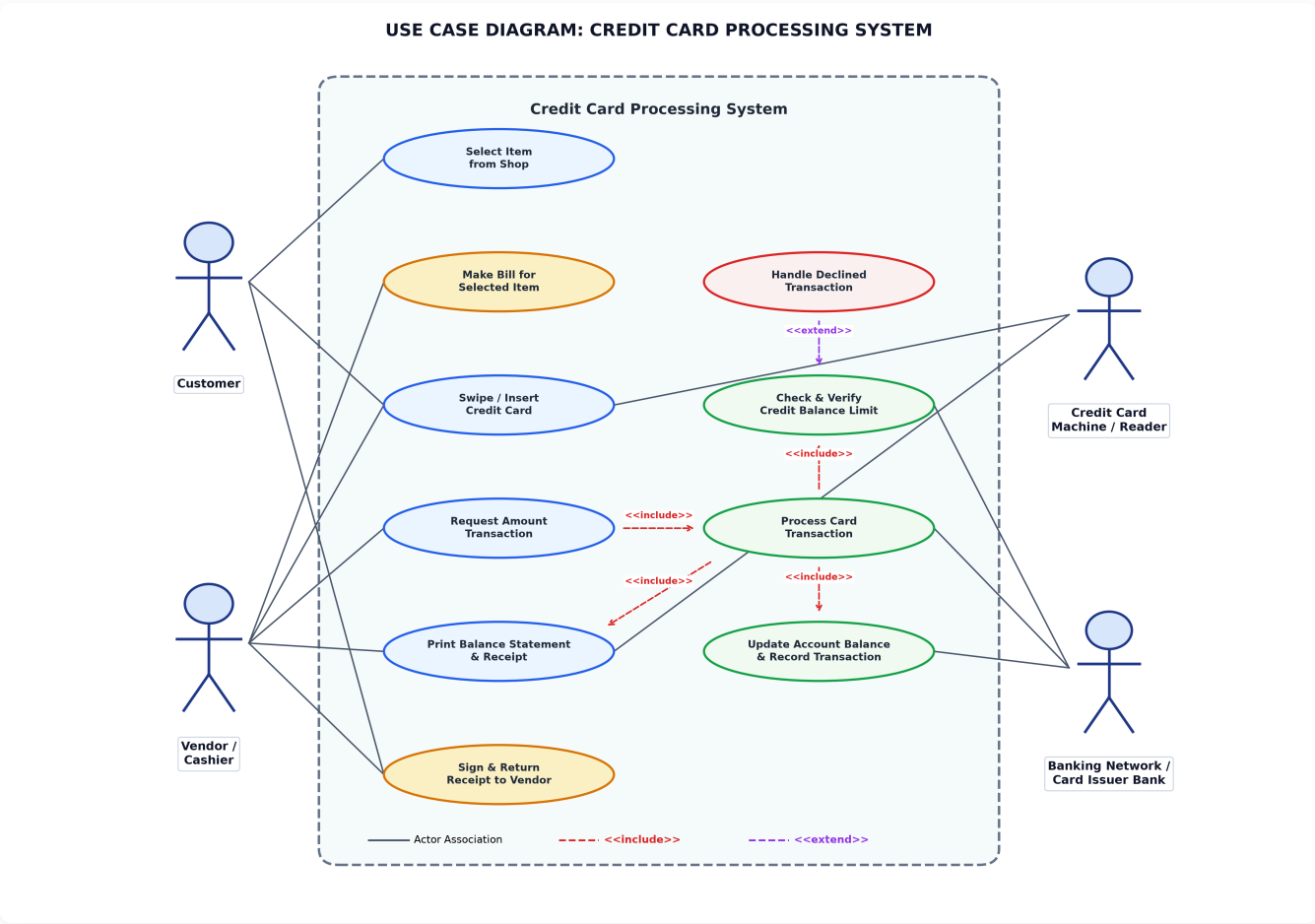


Figure 2.1: Use Case Diagram for Credit Card Processing System (Case Study)

2.4 Actor-to-Use-Case Traceability Matrix

Use Case	Customer	Vendor	Card Machine	Banking Network
Select Item from Shop	✓			
Make Bill for Selected Item		✓		
Swipe / Insert Credit Card	✓	✓	✓	
Request Amount Transaction		✓	✓	
Process Card Transaction			✓	✓
Check & Verify Credit Balance Limit				✓

Use Case	Customer	Vendor	Card Machine	Banking Network
Update Account Balance & Record Transaction				✓
Handle Declined Transaction	✓	✓	✓	✓
Print Balance Statement & Receipt		✓	✓	
Sign & Return Receipt to Vendor	✓	✓		

2.5 Detailed Use Case Specification: Process Card Transaction

Use Case ID & Name	**CC-UC-05: Process Card Transaction**
Primary Actors	Vendor, Credit Card Machine / Reader, Banking Network
Pre-conditions	Bill generated, card successfully swiped, and amount submitted on terminal.
Post-conditions	Account ledger updated, transaction authorization code issued, balance receipt printed.
Main Success Scenario	1. Card reader transmits encrypted card details and amount to bank host. 2. Banking system invokes *Check & Verify Credit Balance Limit* (<<include>>). 3. Bank confirms card is valid and requested amount is within allowable credit balance limit. 4. Bank invokes *Update Account Balance & Record Transaction* (<<include>>). 5. Bank sends approval confirmation to card machine. 6. Card machine invokes *Print Balance Statement & Receipt* (<<include>>).
Alternative Flows	**3a. Balance Exceeded:** If bill exceeds credit limit, bank returns rejection token; system executes *Handle Declined Transaction* (<<extend>>). No funds are debited.